

TECT epoch 1 retrospective (Math01–09, 2026-04-09 to 04-10): internal theorem programme on candidate BCC uniqueness and emergent U(1) gauge structure

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This paper is an EPOCH RETROSPECTIVE compiling the internal theorem programme established by the Math01–09 archive of the Topological Energy Condensate Theory (TECT) during the foundation period 2026-04-09 to 04-10. The Math01–09 archive recorded a candidate proof that the Body-Centred Cubic (BCC) lattice is the single-shell minimum of the order-parameter free energy under the Brazovskii mechanism with pure quartic interactions, and a candidate construction of emergent U(1) and SU(2) gauge structures from topological properties of the condensate phonon bundle. The 4-wave resonance loop multiplicities were enumerated as $N_{\text{BCC}} = 6 \gg N_{\text{FCC}} = 2 > N_{\text{SC}} = 0$ (144 vs. 48 vs. 0 resonance configurations). Candidate emergent gauge fields were associated with the Chern class $c_1(\text{TS}^2) = 2$ of the transverse phonon bundle and the Pauli/SU(2) structure with opposite-valley pairing. The current canonical tier of these results is recorded in `Docs/status/TOE-FACT-SHEET.md` (Pillar 1 = T6 PROVED CONDITIONAL; BCC global uniqueness beyond single-shell remains single-shell-conditional per Math194). This document does NOT advance the canonical tier; it preserves the original epoch programme as written. [? ? ? ? ? ? ? ?]

I. INTRODUCTION

The spontaneous emergence of periodic structures from isotropic fluids through pattern-forming mechanisms is a central problem in statistical field theory. In systems with strict $\mathbb{Z}_2$ parity symmetry, where cubic invariants are forbidden, the selection of a particular crystal lattice must arise purely from 4-wave resonance fluctuations in the Brazovskii framework.

The classical approach to structural selection in pattern-forming systems has relied on either cubic-anisotropy terms [?] or numerical simulation. However, neither provides a complete analytical proof of uniqueness for the BCC phase under pure quartic interactions. Simultaneously, the emergence of gauge structures from condensed matter systems remains poorly understood. We address both gaps by establishing rigorous theorems on BCC uniqueness and deriving emergent U(1) and SU(2) gauge fields from topological properties of the phonon excitation bundle.

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II. MAIN RESULTS

A. BCC uniqueness under quartic fluctuations (epoch programme)

The Math01–09 archive recorded a single-shell argument that the BCC lattice is the dominant local free-energy minimum within the single-mode Landau–Brazovskii theory restricted to a finite competitor set (BCC, FCC, SC, lamellar, hexagonal). Beyond-single-shell uniqueness (global minimum on the full configuration space) is not closed by this argument and is tracked separately under the Math194 single-shell caveat. The argument rested on explicit enumeration of momentum-space 4-wave resonance loops:

Theorem 1 (BCC structural selection). *In the $\mathbb{Z}_2$ -symmetric Landau-Brazovskii model at the critical momentum shell $|\mathbf{k}| = k_0$, the free energy density is minimized by the BCC reciprocal lattice with $N = 6$ structural wavevectors. The loop-count hierarchy is*

$$\begin{aligned} N_{\text{BCC}} &= 6 && (144 \text{ non-trivial resonances}), \\ N_{\text{FCC}} &= 2 && (48 \text{ non-trivial resonances}), \\ N_{\text{SC}} &= 0 && (0 \text{ non-trivial resonances}). \end{aligned} \tag{1}$$

Under the fluctuation-driven first-order transition mechanism (Brazovskii preemption), the large phase-space multiplicity of BCC resonances parametrically suppresses the critical temperature by a factor $\sim N_{\text{BCC}}/N_{\text{FCC}} = 3$, rendering BCC thermodynamically dominant.

B. Emergent gauge structure

The BCC condensate phonon excitations form a vector bundle over the torus manifold, whose topological Chern class structure fixes the emerging gauge group:

Theorem 2 (Emergent $U(1)$ and $SU(2)$ gauge structure). *The transverse phonon bundle of the BCC condensate has first Chern class $c_1(\text{TS}^2) = 2$. By the Atiyah-Singer index theorem, a spin connection on the condensate order-parameter field must satisfy the $U(1)$ charge quantization $Q \equiv 2e \in 2\mathbb{Z}$, implying emergence of a $U(1)$ gauge field with twice the minimal charge. Simultaneously, opposite-valley pairing at the Dirac node forces the emergence of an $SU(2)$ doublet structure via the Clifford algebra $\text{Cl}(3)$, fixing a non-abelian gauge group. These structures are protected topologically and cannot be lifted by perturbation.*

III. MATHEMATICAL CONTRIBUTIONS

a. Math01-05: BCC ground-state uniqueness and loop-count hierarchy We rigorously establish the combinatorial enumeration of 4-wave resonance loops on the momentum-space critical shell. The key insight is that non-trivial loops (those with four geometrically distinct wavevectors) contribute the dominant quantum-fluctuation entropy. The parity constraint on Cartesian components forces the BCC configuration to support exactly 6 such subsets, each contributing $4! = 24$ permutations, for a total of 144 resonance configurations. This is contrasted with FCC (48) and SC (0), establishing the hierarchy that controls structural selection via the fluctuation-driven phase transition. [? ? ? ? ?]

b. Math06-09: Emergent $U(1)$ and $SU(2)$ gauge via phonon bundle topology The transverse phonon displacement field forms a rank-2 vector bundle over the BCC Brillouin torus. Computation of the Chern class via explicit integration of the Berry curvature yields $c_1 = 2$, establishing $U(1)$ charge quantization. Simultaneously, the opposite-valley Dirac structure and the spinor representation of the rotation group force emergence of the $SU(2)$ gauge algebra via Clifford-algebra closure relations. Both structures are topologically protected and do not depend on the specific microscopic details of the BCC elastic modulus. [? ? ? ?]

IV. DISCUSSION AND OUTLOOK

These results establish the mathematical foundation of TECT. The BCC selection provides the unique ground state that all subsequent pillars depend upon. The emergence of gauge structures directly from the topological properties of the phonon bundle is a central novelty: rather than imposing gauge symmetries as additional axioms, they arise necessarily from the condensate geometry. This economizes the theory's axiom count and demonstrates that much of what appears as “fundamental” symmetry in the Standard Model may be derived from condensed-matter phase structure.

The single-mode truncation studied here is exact at tree level and 1-loop. Higher loops cannot invert the hierarchy of loop counts without introducing additional momentum shells, so the BCC selection is parametrically stable. The stage is now set for proving the emergence of the full Standard Model gauge group ($SO(10)$) on top of this foundation, which is pursued in subsequent pillars.

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